Examiner only

5. Bethan investigates damped oscillations and notices that the amplitude of an oscillating system decreases from 1.5 ± 0.1 cm to 0.5 ± 0.1 cm in 12 minutes.

(a) Suggest what causes the decrease in amplitude. [1]

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(b) This reduction in amplitude may be represented by:

$$A = A_0 e^{-\lambda t}$$

where A is the amplitude at time t, A_0 is the initial amplitude, and λ is a characteristic constant for the system.
By determining the maximum and minimum possible values of λ, determine its value and its **absolute** uncertainty. [5]

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(c) Give **one** example of damped motion that occurs in everyday life. [1]

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